

Experiment No. 7

Aim: Malware Analysis of various files types(Java Script, pdf)

Title: Malware Analysis of various files types using various tools.

Part A:

The MS-Office documents had two types of structures

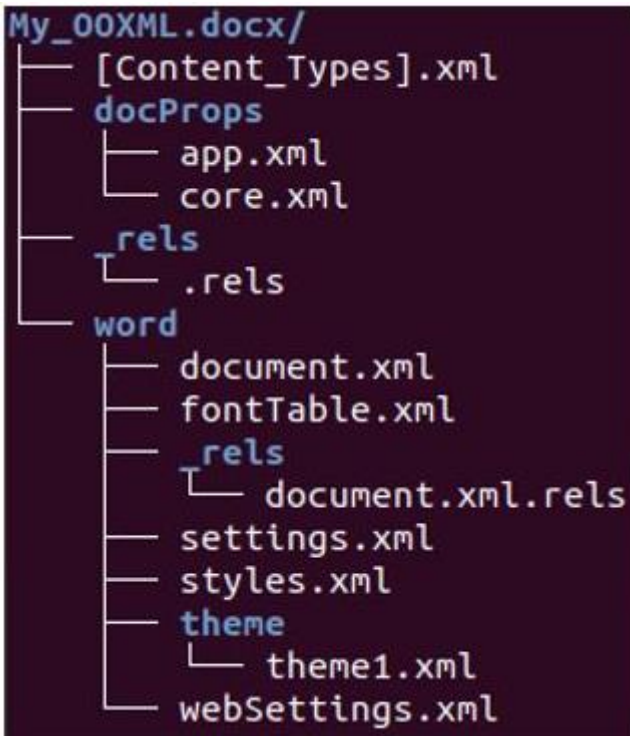
OOXML (Newer Version)

OLE Compound File (Older Version)

What is Microsoft OOXML?

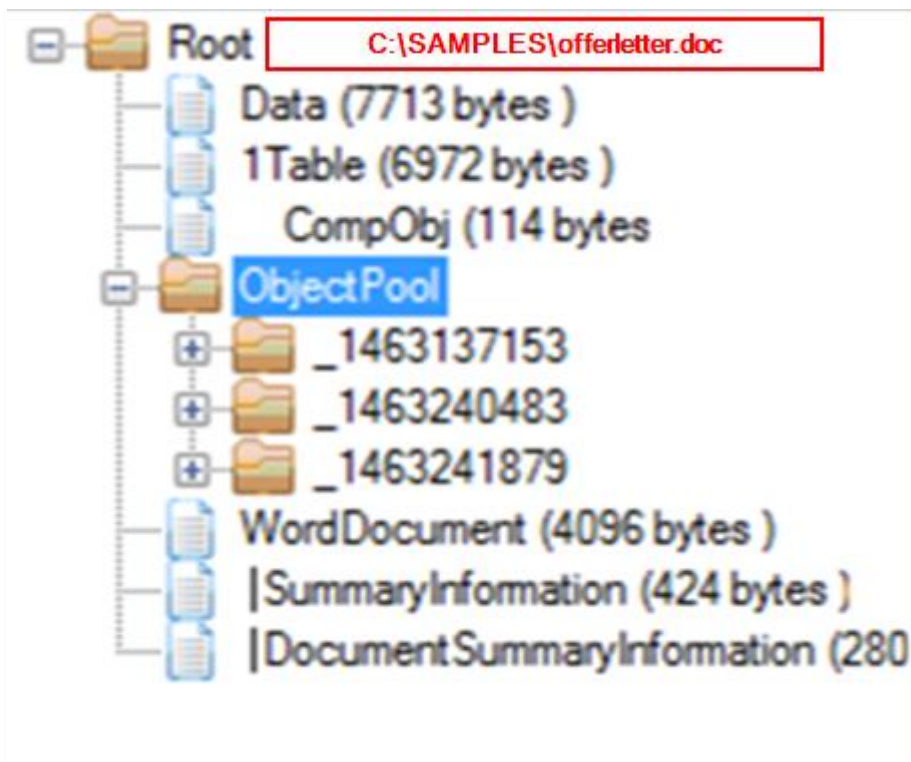
MS-OOXML is ZIP/XML based format from 2007 version of office documents to onwards, developed by Microsoft. It covers word processing documents, spreadsheets, presentations, charts, diagrams, shapes, and other graphical material.

OOXML files are ZIP archives containing various XML files (parts) and organized into a single package. This breaking up or chunking of the data into pieces makes it easier and quicker to access data and reduces the chances of data corruption. The parts can contain any type of data; to keep track of the data type of each part without relying on file extensions, the type for each part is specified in a file within the package called [Content_Types].xml.



What is OLE Compound File

OLE is a mechanism that allows users to create and edit documents containing items or “objects” created by multiple applications. OLE was originally an acronym for Object Linking and Embedding.



In the OLE Types documents, there is root storage(main file) that contains different streamed and storage objects, and then those objects and also contain different streamed and storage objects.

Part B:

1. For Java Script analysis:
 - a. <https://dhanushnehrumedium.com/javascript-deobfuscation-7da22f0b98fa#:~:text=Code%20deobfuscation%20is%20an%20important%20skill%20to%20learn%20if%20we,though%20performance%20may%20be%20slower.>
 - b. <https://blog.jscribler.com/javascript-obfuscation-the-definitive-guide>
2. For PDF analysis:
 - a. <https://www.intezer.com/blog/incident-response/analyze-malicious-pdf-files/>
 - b. <https://zeltser.com/tools-for-malicious-pdf-analysis/>
3. List of malicious pdf files:
 - a. <https://github.com/luigigubello/PayloadsAllThePDFs/tree/main/PDF%20Files>
4. For rtf file analysis:
 - a. <https://foxptr.medium.com/analyzing-rtf-documents-5bb45071adfd>
5. Conclude with output screenshots.